Effect of CH_4 Plasma Pretreatment on Cu-to-Cu Bonding Applications

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***Abstract*— Cu-to-Cu direct bonding offers significant advantages for advanced 3D integration, including reduced power consumption and a lower thermal budget. In this study, a reductive plasma-based Cu-to-Cu direct bonding method is proposed for stacked high bandwidth memory (HBM) applications. While conventional Ar/H_2 (5%) forming gas plasma is commonly used for Cu oxide reduction, its low hydrogen radical and ion density results in extended treatment times and elevated temperatures. To address these limitations, an Ar/CH_4-based plasma process is introduced, providing a higher density of hydrogen radicals and ions, thereby enhancing the reduction efficiency under low temperature conditions.**

An 8-inch silicon wafer with a Ti/Cu metal layers was fabricated using DC magnetron sputtering, followed by Cu/barrier chemical mechanical polishing (CMP) and CH_4 plasma treatment via inductively coupled plasma chemical vapor deposition (ICP-CVD). The Cu-to-Cu bonding process was carried out for 1 hour at 260 °C, and this was followed by a subsequent annealing step at 200 °C to promote Cu–Cu bonding through grain boundary diffusion. Plasma process parameters including power, gas flow rate, and pressure were optimized for Cu oxide removal and surface passivation.

X-ray photoelectron spectroscopy (XPS) and transmission electron microscopy (TEM) analysis confirm that Ar/CH_4 plasma effectively reduces Cu oxides. Furthermore, a carbonaceous layer with a thickness of 1–3 nm was formed on the reduced Cu, acting as a passivation layer to suppress reoxidation. Scanning electron microscopy (SEM) analysis of the post-bonded structure revealed a void-free bonding interface, demonstrating the effectiveness of the proposed method in achieving high-quality Cu-to-Cu direct bonding.

Keywords*— *Hybrid Cu bonding, Reducing Plasma pre-treatment, Reduction of Cu oxide, Passivation of metallic Cu surface, Plasma pretreatment

I. INTRODUCTION

As chip stacking technology continues to advance, power consumption and thermal management have emerged as critical challenges in three-dimensional (3D) integration. One promising solution to mitigate these issues is the reduction of bump pitch size to the sub-micron level, thereby minimizing interconnect parasitics and improving signal integrity. However, conventional solder bump technology is fundamentally limited in achieving sub-micron pitch sizes due to its reliance on surface tension and reflow processes [1-5].

To overcome this limitation, Cu-to-Cu direct bonding has become indispensable, enabling fine-pitch interconnections without the need for solder materials. Despite its advantages, Cu-to-Cu direct bonding still faces significant challenges. Cu_2O (Cuprous oxides) and CuO (cupric oxides), which are the native oxides on Cu surface, severely degrades bonding quality by hindering atomic diffusion across the bonding interface. Moreover, even after initial oxide removal, the reduced Cu surface remains highly reactive and prone to reoxidation under ambient conditions [2].

To address these challenges, various surface pretreatment techniques have been proposed, particularly plasma-based reduction methods [1-6]. Among them, Ar/H_2 (5% H_2) forming gas plasma has been widely used, as hydrogen radicals and ions within the plasma are known to reduce copper oxides effectively while simultaneously cleaning the Cu surface [7,8]. However, the relatively low concentration of hydrogen radicals in conventional Ar/H_2 plasma results in long treatment durations and requires higher process temperatures to achieve sufficient reduction, which increases the thermal budget of the process.

To address these limitations, this study proposes an Ar/CH_4-based reducing plasma as an alternative pretreatment method for Cu-to-Cu direct bonding. CH_4-based plasmas are capable of generating a higher density of hydrogen radicals and ions compared to H_2-based plasmas, thereby improving the reduction efficiency. The Cu oxide reduction mechanism using CH_4 plasma can be described by the following reactions [9]:

$$4Cu_2O + CH_4 \rightarrow CO_2 + 2H_2O + 8Cu \quad (1)$$

$$3Cu_2O + CH_4 \rightarrow CO + 2H_2O + 6Cu \quad (2)$$

In addition, the Ar/CH_4-based plasma generates a high concentration of hydrogen species, resulting in a C_xH_y layer with a correspondingly high hydrogen content. This hydrogen-rich carbonaceous layer exhibits a relatively soft and compliant nature, which facilitates intimate contact at the bonding interface and thereby contributes to improved bonding quality [10].

The CH_4 plasma treatment serves two key functions: (i) reducing Cu oxides through interactions with hydrogen radicals, hydrogen ions, and excited H_2 molecules, and (ii) passivating the reduced Cu surface by forming a protective C_xH_y layer.

II. Methods

A. Experiment

The experimental procedure is illustrated in Figure 1. An 8-inch notch-type silicon wafer was used as the substrate, pre-coated with a 700 nm SiO_2 insulating layer. To promote adhesion between the substrate and metal film, a 50 nm titanium (Ti) layer was deposited, followed by a 1 µm copper (Cu) thin film using DC magnetron sputtering. The sputtering process was conducted under a working pressure of 5 mTorr with a DC power of 2000 W, and an argon (Ar) gas flow rate of 100 sccm.

Subsequent to the metal deposition, a Cu/barrier chemical mechanical polishing (CMP) process was performed to planarize the wafer surface. Thereafter, CH_4 plasma treatment was conducted using an inductively coupled plasma chemical vapor deposition (ICP-CVD) system (Oxford PP100 Cobra300, S-Fab@SeoulTech). During this treatment, the plasma processing time was fixed, while the input power, CH_4 gas flow rate, and chamber pressure were systematically varied to study their effects. The optimized CH_4 plasma conditions derived from this experiment are summarized in Table 1.

Cu-to-Cu direct bonding was performed in a vacuum environment at 260 °C for 1 hour, followed by post-bonding annealing at 200 °C for an additional hour to enhance the interfacial bonding quality.

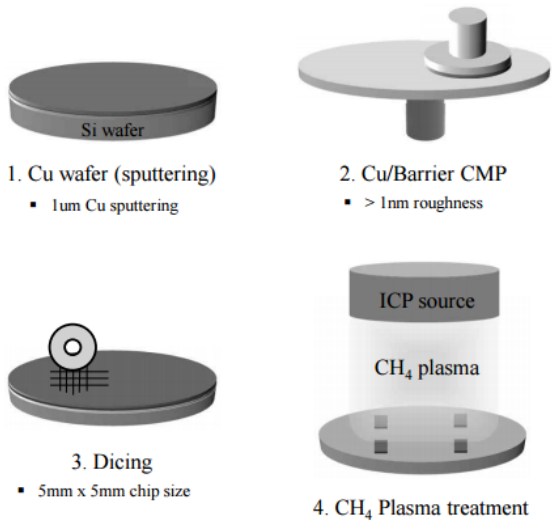


Fig. 1. CH_4 plasma treatment process flow

Table. 1. Process parameters

ICP power	CH_4 gas flow rate	Pressure
[300:2000] (W)	[15:80] sccm	[3:10] mtorr

B. Analysis

The resulting interfacial characteristics were analyzed through X-ray photoelectron spectroscopy (XPS), raman spectroscopy, transmission electron microscopy (TEM), and scanning electron microscopy (SEM) to verify the reduction performance and bonding integrity of the proposed method.

The presence of an a-C:H layer on the Cu surface may deteriorate the electrical conductivity at the metal-metal interface. To evaluate this effect, sheet resistance measurement were performed using a four-point probe, as presented in Figure 2.

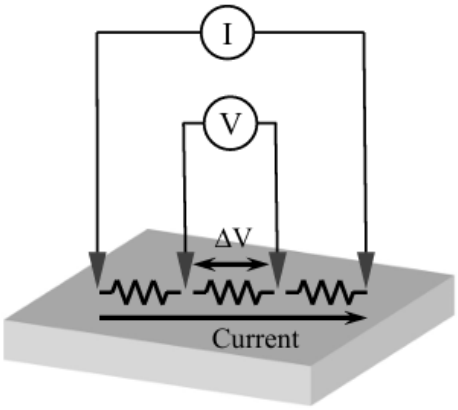


Fig. 2. 4 point probe

III. Results and Discussion

A. DOE & Random Forest model optimization

Design of Experiments (DOE) was employed to model the process using a second-order polynomial regression, as shown in the following equation:

$$Y=\beta_0 + \sum \beta_i X_i + \sum \beta_{ii} X_i^2 + \sum \beta_{ij} X_i X_j + \varepsilon \qquad (3)$$

Where Y is the predicted response variable, $\beta_0, \beta_i, \beta_{ii}, and\ \beta_{ij}$ are the regression coefficients; and ε is the random error term.

In parallel, a Random Forest model, tree-structured machine learning method, was utilized as a data-driven, black-box approach to compare predictive performance as shown in Figure 2.



Fig. 2. Black-box model

To optimize the surface conditions, the extent of Cu oxide reduction and the formation of hydrogen-rich carbonaceous layers on the Cu surface were quantitatively analyzed using X-ray photoelectron spectroscopy (XPS). Both DOE and Random Forest modeling were conducted using three key process parameters: ICP power, CH_4 flow rate, and chamber pressure. Notably, both methods yielded consistent trends, indicating that higher ICP power and increased CH_4 flow rate, combined with lower pressure, are favorable for effective Cu surface activation and passivation. The DOE results are presented in Figure 3 while the Random Forest model output is shown in Figure 4. The final process condition was selected as Case 16 in Table 3, based on its superior performance in achieving the primary objectives—maximum removal of the native oxide layer (Cu_2O) and the highest passivation layer of the carbon film.

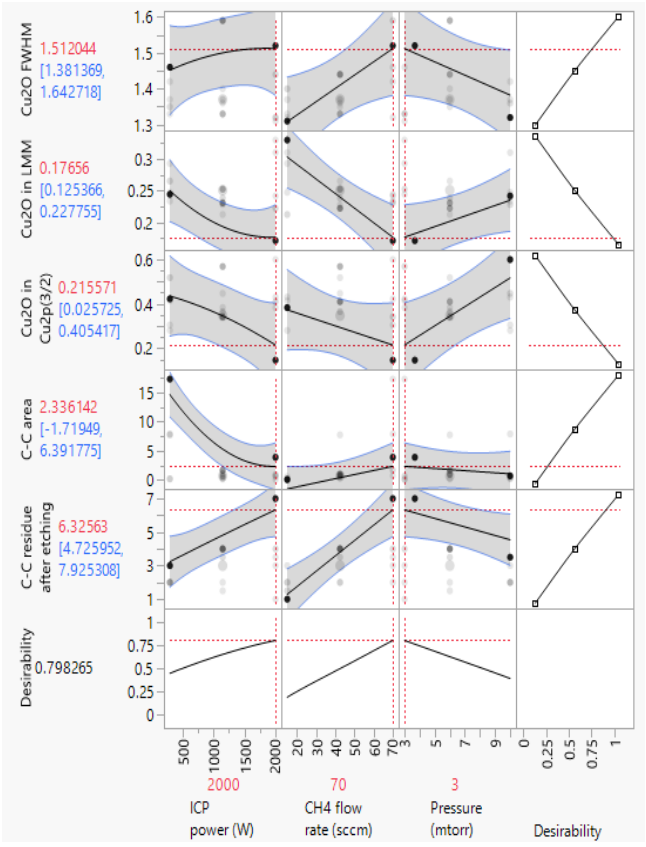


Fig. 3. DOE result

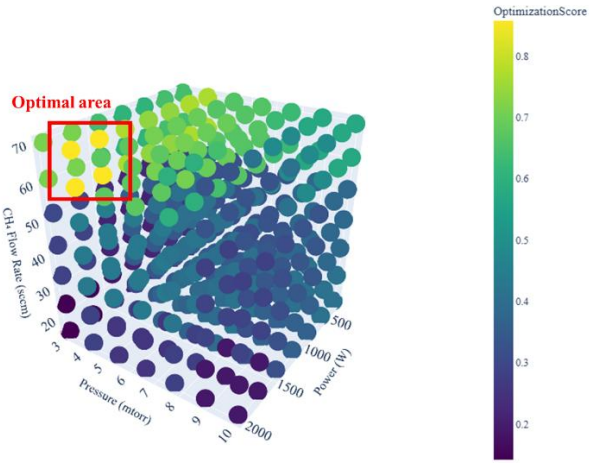


Fig. 4. Random Forest optimization result

Table. 3. Process parameters for 5 selected cases

Case	ICP power	CH_4 gas flow rate	Pressure
6	1150 W	42.5 sccm	6 mTorr
9	300 W	70 sccm	10 mTorr
11	300 W	70 sccm	3 mTorr
12	300 W	15 sccm	6 mTorr
16	2000 W	70 sccm	3.7 mTorr

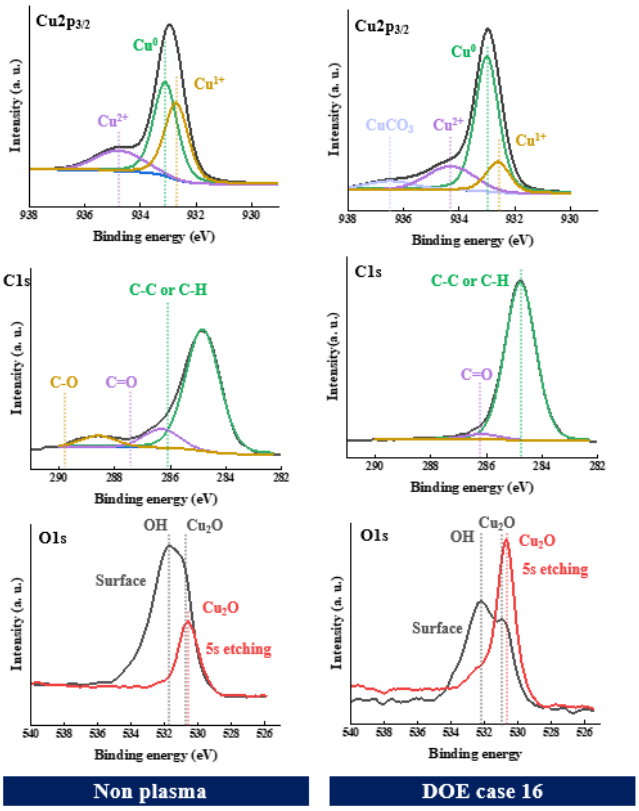


Fig. 5. XPS results of non plasma and case 16

A-1) Chemical analysis by XPS

Cu2p : Following CH_4 plasma treatment, Cu^+ is reduced to metallic Cu, confirmed by $Cu2p_{1/2}$ deconvolution.

C1s : Post-treatment spectra reveal a notable enhancement of C–C/C–H bonding, indicative of surface passivation. In contrast, the disappearance of C=O and C–O components confirms effective removal of surface oxides and contaminants.

O1s : Surface Cu_2O decreases and OH increases after plasma treatment, suggesting chemical reduction and surface activation. This trend is clearly confirmed by the 5-second etching profile.

A-2) Structural analysis by TEM

Prior to plasma treatment, a native oxide layer with a thickness of approximately 1-2nm was formed on the Cu surface. However, following the CH_4 plasma treatment, the native oxide layer was effectively removed, and a hydride carbonaceous layer (a-C:H) with a thickness of approximately 2nm was deposited.

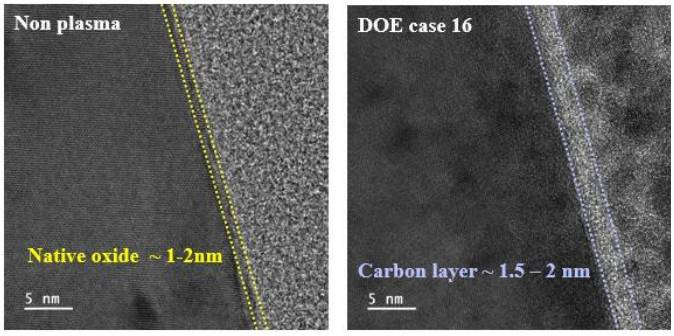


Fig. 6. TEM results of non plasma and case 16

B. Resistivity analysis by four-point probe

As shown in Figure 7, the resistivity values derived from sheet resistance measurements taken immediately after plasma treatment indicate that the reductive plasma reduces resistivity by removing the native oxide layer; however, the reduction is relatively modest. The process parameters for the selected cases are summarized in Table 3. In these cases, effective oxide removal was achieved, and a hydrogenated amorphous carbon (a-C:H) layer with a thickness greater than 1 nm was deposited.

The presence of the a-C:H layer appears to have a negligible impact on the measured resistivity, likely due to its limited thickness (1–3 nm). However, it should be noted that four-point probe measurements may not accurately reflect the resistivity of ultrathin films in this range, as the probe tips can penetrate beyond the ultra thin film, thereby skewing the results slightly. To address this limitation, contact resistance measurements after bonding are planned as part of subsequent evaluations.

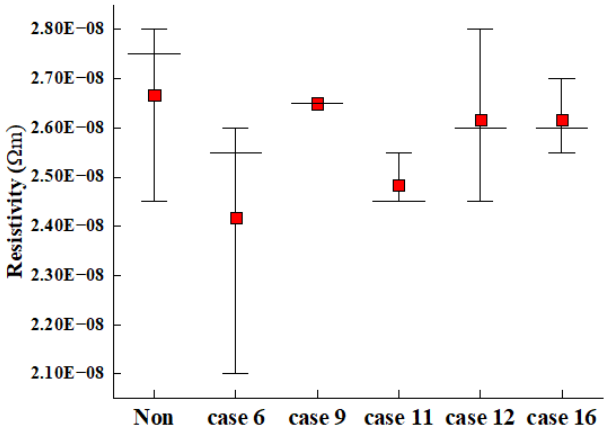


Fig 7. Resistivity(Ω·m) of Cu surface

C. Bonding interface analysis by SAT and SEM

As depicted in Fiture 8, the SAT and SEM analysis reveal that a hydride carbonaceous layer was formed after CH_4 plasma treatment, and no voids were observed ar the bonding interface. This indicates that the presence of the a-C:H layer does not interfere with the bonding process.

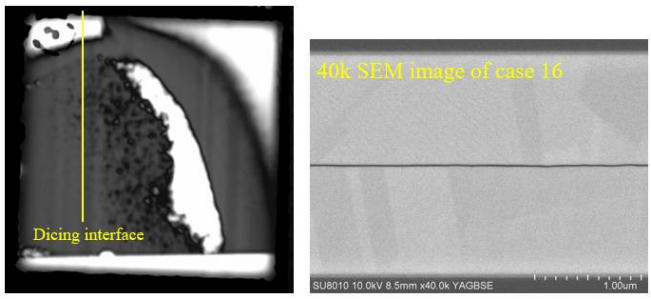


Fig. 8. SAT & SEM image of case 16

IV. Summary

To improve Cu-to-Cu bonding, this study examines how CH_4 plasma pretreatment modifies the Cu surface. Surface analyses confirm that CH_4 plasma effectively reduces native copper oxides. The reduction efficiency increases with higher plasma power and lower pressure, suggesting that both high-energy hydrogen radicals and ions play a critical role in promoting low-temperature reduction.

Additionally, the CH_4 plasma induces the formation of a hydride carbonaceous layer (a-C:H) on the Cu surface, which passivates the exposed metal and prevents reoxidation in air. The physical properties of the a-C:H layer vary depending on plasma conditions: at high plasma power, the layer becomes denser and more tightly bonded, whereas low-power plasma results in a softer, loosely aggregated morphology. Further investigation will be conducted to determine which type of a-C:H layer is more advantageous for the low temperature bonding process.

In conclusion, CH_4 plasma pretreatment imparts two key effects on the Cu surface: (1) it removes native oxides, thereby enhancing Cu diffusion during bonding, and (2) it forms an a-C:H layer that mitigates reoxidation under ambient conditions. These dual functions are expected to significantly improve Cu-to-Cu bonding quality and enable lower-temperature bonding processes.

ACKNOWLEDGMENT

This research was conducted with the support of the Technology Innovation Program (Public-Private Joint Investment Semiconductor R&D Program [K-CHIPS] aimed at fostering advanced professionals; RS-2023-00236091, Development of hybrid bonding technologies for 3D packaging interconnects), funded by the Ministry of Trade, Industry and Energy (MOTIE, Republic of Korea; 1415187584). Additional support provided by a grant of the Basic Research Program funded by the Korea Institute of Machinery and Materials (grant number : NK254A, Development of Core Technologies for Advanced Chjplet Packaging Equipment).

The authors also acknowledge the use of the ICPCVD system (PP100 Cobra300 ICPCVD, Oxford) at S-Fab, Seoul National University of Science and Technology, which was utilized for CH_4 plasma treatment experiments in this study.

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